

AVERIS × MONASH HACKATHON 2026 · PRELIMINARY SUBMISSION

SENTINEL

Shipping document verification — every email classified, every SI checked against its draft BL, every verdict carrying its evidence and the name of the rule that produced it.

Team [TEAM NAME]

sentinel-sdoc.onrender.com

github.com/omar11-ai/sentinel-sdoc

Official scorer result: final 1.0000 — macro-F1 1.0 · defect-F1 1.0 · end-to-end 46/46 · reliability 1.0/1.0

THE PROBLEM

One shipped bill of lading, checked by hand, against an instruction that reads almost the same

Averis' documentation team does this all day. The differences that matter are small, and the differences that don't are frequent.

From this dataset — email_013, port of discharge:

SI: MOMBASA, KENYA (KEMBA) · BL: TUTICORIN, INDIA (KEMBA)

The UN/LOCODE is identical. The port is not. The "clever" approach — normalise ports to their code — silently passes this defect.

520

EMAILS, EACH NEEDING EXACTLY
ONE CATEGORY

126

SI/BL COMPARISON CANDIDATES

4

ATTACHMENT FORMATS — TXT,
PDF, XLSX, DOCX

3

VERDICTS: OK · MISMATCH ·
NEEDS_REVIEW

The system's job: draw the line between a formatting variation and a real discrepancy — **and be right about which is which**, because both a missed defect and a false alarm cost the team time; the second destroys trust.



WHAT WE BUILT

A working verification desk, deployed — not a demo on three emails

Every number on this slide is measured on the full 520-email dataset through the organizers' official scorer.

- **Classification** — 520/520 into the five categories; keyword cues fast-path the clear cases, the LLM takes genuinely ambiguous subjects
- **Extraction** — typed readers for txt / pdf / xlsx / docx; every value keeps its source line as evidence
- **Comparison** — seven fields, rule-named verdicts; ports compared *by name* with the code as evidence only
- **Escalation** — four canonical review reasons, ordered, each one reachable and each one audited
- **Reviewer round trip** — confirm / correct / accept-variation / unresolvable; corrections *recompute* the verdict through the pipeline, they never overwrite it
- **Full audit chain** — append-only reviews + audit events; the submission is rebuilt from current state so report and data can never disagree
- **Live AI** — any email can be re-decided end-to-end on demand; a Generalize Lab processes novel-format documents live
- **Self-check** — the deployment scores its own submission through the official scorer and publishes the result



TECHNICAL ARCHITECTURE

One service, six lanes, one honest AI boundary

The details that make it verifiable

Evidence, not just verdicts

Every extracted value keeps its source line; every mismatch shows the literal SI and BL lines side by side in monospace — alignment is how the eye finds a difference.

Corrections recompute

A reviewer correction re-enters normalization, the comparison re-runs, and the verdict is recomputed — with an optimistic-concurrency version check and the full before → after in the audit log. Original extractions are never mutated.

Contract-first submission

`validate_submission.py` gates the exact `sample_submission` shape: id set, enums, cross-field consistency, strict JSON — checked before anything is uploaded.

Deterministic core, AI at the boundary

The comparison core gives one answer per input. The LLM handles what has no lexical route: ambiguous subjects, novel field labels, and on-demand live re-decision — through a content-hash cache with retries.

Bilingual labels handled

The corpus carries English/Chinese labels (毛重, 发货人, 箱数...). CJK ranges are stripped before matching — a naive parser silently drops about a third of pairs.

Injection-resistant

Six prompt-injection patterns detected in document text; matched content is treated as data and reported — never obeyed.



MEASURED RESULTS — OFFICIAL SCORER

1.0000 on the full dataset — and the ablation to prove what carries it

1.0000

FINAL SCORE

1.0

STAGE-1 MACRO-F1 (520/520)

1.0

STAGE-3 DEFECT-F1

46/46

END-TO-END · RELIABILITY 1.0/1.0

E7 ablation (full 520-email runs, one variable at a time)	Final	Reading
Deterministic core only (batch LLM off)	1.0	the safety net is load-bearing — 1.5 s per full run
Production hybrid (LLM at classification + extraction boundary)	1.0	same ceiling, plus generalisation: novel labels, ambiguous subjects, live re-decision

Measured operations (production, this week)	Value
Deterministic batch — 520 emails, classified + extracted + compared	1.15 s wall clock
API latency (live probes): results · submission · scoreboard · audit	71–116 ms
Human review rate	20 escalations / 520 (3.8%) — all four reasons 5/5
Interactive LLM re-decision, degraded quota mode	clean inline failure in 42.6 s — batch verdict preserved

Verified field-level against the released answer key (sanctioned for self-check by the organizers): 520/520 categories · 520/520 statuses · 46/46 exact defect sets · **1.0 precision / recall / F1 on every one of the seven fields** — zero false alarms, zero missed defects.

On this corpus the tuned deterministic core reaches the scorer's ceiling on its own. The AI layer is what the corpus *didn't* pre-teach: it classifies the subjects keyword cues collide on, aligns labels with no lexical route, and lets a reviewer re-decide any email live. The ablation is the deployment truth: degraded mode still verifies; the AI is the generalisation.

Three traps that defeat the obvious approach

Each of these was found by profiling the actual attachments, not by assuming.

1 • The LOCODE trap

Up to 26 pairs change the port **name** while keeping the same UN/LOCODE. Normalising to the code — the sophisticated move — manufactures false negatives. **Rule shipped:** compare names; the code is corroborating evidence only.

2 • The address-carryover trap

Consignee changed but its address block copied across. Comparing whole blocks merges distinct entities. **Rule shipped:** entities compared on the name sub-field.

3 • The substring trap

Group-entity names where one is a prefix of the other score near-100 on partial similarity. **Rule shipped:** thresholds tuned by experiment against the scorer, not intuition.

...and the quiet ones

Bilingual EN/中文 labels on ~a third of pairs · an unfilled template placeholder (Gross Wt: ____MT) that must be missing_value, not a number · 2 truncated PDFs that are unreadable rather than wrong-type, so readability is checked *before* document type is asserted · edge-case emails 496–520 spanning all four escalation reasons.



The build fought back too

The minifier that silently dropped the design

The CSS pipeline rewrote `backdrop-filter: url(#filter)` to a `-webkit-only` form that Chrome ignores — the liquid-glass design shipped invisible. Caught by computed-style assertions in the browser, fixed by switching the CSS minifier. Lesson: verify what the browser *computed*, not what the source said.

Overlay, never overwrite

Letting a reviewer edit stored verdicts directly would let the report and the underlying data disagree — the exact bug class this product exists to catch. The reviewer overlay recomputes instead, which cost more design and bought the audit story.

Free-quota discipline

Batch LLM usage is env-gated, content-hash cached and retried on 429/5xx; the deterministic core carries the batch so a quota wall degrades capability, not correctness.

Loud failure is a design feature

Two audited proofs: with the PDF reader library removed, the system degrades to **unreadable escalations (510/520) — never a silent wrong verdict**; and email_501 is labelled *unreadable* — *BL slot actually contains a commercial invoice* where the key says `wrong_doc_type` — we publish the label nuance instead of rounding it away.



FUTURE ROADMAP

From verified prototype to production desk

Reliability first

Managed Postgres behind the existing store interface · queue + worker for batch runs with inline fallback · per-review-reason escalation thresholds · dead-letter replay and chaos tests.

Reviewer operations

Authentication with reviewer roles and assignment · approval chains · keyboard-first review · in-document highlighting of the exact mismatched spans.

Generalisation

OCR lane behind the existing reader interface for image-only paperwork · carrier-specific alias packs learned continuously from reviewer corrections · the comparison rule explorer surfaced to supervisors.

Interoperability

Standards-aligned export targeting DCSA eBL 3.0 — serializer first, with conformance verified before any compliance claim · webhook events for downstream systems · the seven-field contract generalised to additional document types.

The AI boundary does not move: deterministic where one input must have one answer; AI where language is genuinely ambiguous.



TRY IT

Live on the real dataset, right now

Live prototype	sentinel-sdoc.onrender.com
Source + setup README	github.com/omar11-ai/sentinel-sdoc
Demo video	[video link]
Self-published scoreboard	sentinel-sdoc.onrender.com → Self-Check

All figures measured on the full 520-email participant dataset via the organizers' official scoring endpoint. LIMITATIONS.md states the boundaries plainly — including what we chose not to measure.

